

Homework assignment for 02YMECH

Chapter: Motion in a Noninertial Reference Frame

1. Check the validity of the phrase “apple doesn’t fall far from the tree” for an apple tree at the equator after considering the Coriolis force. Consider that the apple falls from the height $h = 4$ m. Neglect the air resistance.

Hint: Neglect the effect of the horizontal velocity caused by the Coriolis force: If the falling apple starts with zero horizontal velocity, then its horizontal velocity appears only because of the Coriolis force itself. Strictly speaking, once this horizontal component exists, it should generate a second-order Coriolis correction (because now velocity has a horizontal piece). That correction is extremely small, neglect it.

2. A particle of mass m moves without friction inside a smooth, rigid, straight pipe that is fixed on a horizontal rotating platform. The pipe is oriented radially outward from the axis of rotation. The platform rotates with a constant angular velocity Ω . Initially, the particle is held at radius r_0 with zero velocity relative to the pipe and then released.
 - (a) In the rotating frame of the platform, write down the equation of motion for the particle, including all inertial forces.
 - (b) Solve for $r(t)$.
 - (c) Determine the time it takes for the particle to reach the end of the pipe at radius $R > r_0$.
 - (d) Explain physically why the particle accelerates, even though its velocity is initially zero in the rotating frame.